

Lambert Leong, PhD



Senior Applied Scientist — Healthcare Foundation Models — Clinical GenAI Evaluation

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Experience

Senior AI/ML Applied Scientist

September 2024 – Present

UnitedHealth Group – Optum AI Applied Scientist

Bellevue, WA

- Enabled AI review board approval for 5 patient-affecting clinical GenAI products by designing automated evaluation metrics, calibrated clinical SME review, and cross-functional testing, driving \$1.5M realized benefit
- Advanced healthcare service recommendations in the mobile app by training an award-winning LLaMA-style foundation model with a custom CPT tokenizer for next-code prediction
- Accelerated enterprise HR response design by co-designing an AI-native agentic triage workflow from stakeholder discovery and guiding AI/ML and data science teams as director of development

Postdoctoral Researcher | Affiliated Developer

August 2023 – September 2024

University of Washington

Seattle, WA

- Enabled large-volume CT spleen volumetry analysis for hematologic malignancy treatment response by implementing SAM/TotalSegmentator segmentation workflows, generating model outputs, and performing statistical analysis
- Improved CT cohort discovery by developing hybrid rule-, regex-, and embedding-based NLP case finding with 98% accuracy across clinical text and research databases
- Accelerated CT body-composition analysis 8x by building a multi-threaded, GPU-batched deep learning inference pipeline

Graduate Researcher

November 2018 – May 2023

University of Hawaii Cancer Center

Honolulu, HI

- Improved generative medical image quality 16% by designing a custom biophysics-informed loss function
- Improved downstream whole-body X-ray prediction tasks 15% by implementing self-supervised image embeddings
- Improved breast cancer risk prediction AUCs 10% over published baselines by implementing multi-view parallel siamese learning networks
- Increased cancer prediction AUC 17% by engineering imaging features with spectral decomposition and convex hull clustering
- Produced 2 conference presentations by mentoring undergraduates in ML modeling for disease prediction

Machine Learning Pathway Intern

May 2020 – August 2020

National Aeronautics and Space Administration (NASA)

Houston, TX

- Improved spacesuit ergonomics analysis by fusing exterior 3D body scans with X-ray scans using shape and appearance modeling to visualize joint rotation centers
- Standardized 3D point-cloud alignment by writing and testing a Python implementation of local coordinate-system transforms

Research Scientist

August 2015 – October 2018

Eyegenix LLC

Honolulu, HI

- Supported successful artificial cornea implantation in animal trials by performing data-driven protein polymer design
- Accelerated startup R&D iteration by rapidly prototyping, testing, and communicating technical tradeoffs to stakeholders

Education

PhD - Bioengineering

2023

Advisor: John Shepherd, PhD

GPA: 3.95/4.00

- Dissertation: *Reducing the Burden of Cancer with Artificial Intelligence*

MS - Computer Science

2018

Advisor: Henri Casanova, PhD

GPA: 3.89/4.00

- Thesis: *A Heuristic for Optimizing the Physical Layout and Network Topology of Integrated 3D Multi-chip Systems Under Temperature Constraints*

BS - Biology

2014

Minor: Chemistry

Core Competencies & Skills

- 9 yrs – **Python**, Pandas/NumPy; ownership of research agendas, applied ML projects, and model evaluation design
- 8 yrs – **Deep Learning**: CNNs, ViTs, Transformers; Generative AI (VAEs, Diffusion); **PyTorch, TensorFlow**
- 2 yrs – **NLP/LLMs** (BERT, LLaMA, RAG), clinical GenAI evaluation, **PySpark**/Databricks, and cloud ML workflows
- 1 yr – **Healthcare Foundation Models & Recommender Systems**: autoregressive CPT prediction, search, collaborative filtering

Fellowships

Google Computer Science Research Mentorship Program	2023
NIH Artificial Intelligence/Machine Learning Consortium to Advance Health Equity and Researcher Diversity	2023

Awards

Make IT Happen Award - UnitedHealth Group / Optum, top 80 engineers company-wide	2026
Merit Based Awards for Research - University of Hawaii	2023
Outstanding Dissertation Award - University of Hawaii	2023
Trainee Travel Grant (3 consecutive years) - University of Hawaii Cancer Center	2019 – 2021
Graduate Research Grant - University of Hawaii	2020

Notable Publications

[1] **Leong, Lambert T** et al. Sparseehr: Scalable foundation modeling for structured ehr via conditional computation. In *ICML 2026 Workshop on Spurious Dynamics for Health*, 2026. Accepted.

[2] Kyle A. Hasenstab, Jie Lu, **Leong, Lambert T.**, Emily Bossard, Evye Pylarinou-Sinclair, Karthika Devi, and Guilherme M. Cunha. Relationship between spleen volume and diameter for assessment of response to treatment on ct in patients with hematologic malignancies enrolled in clinical trials. *Abdominal Radiology*, 50:5799–5809, 2025.

[3] **Leong, Lambert T**, Michael C Wong, Yong E Liu, Yannik Glaser, Brandon K Quon, Nisa N Kelly, Devon Cataldi, Peter Sadowski, Steven B Heymsfield, and John A Shepherd. Generative deep learning furthers the understanding of local distributions of fat and muscle on body shape and health using 3d surface scans. *Communications Medicine*, 4(1), 2024.

[4] Yannik Glaser, John Shepherd, **Leong, Lambert**, Thomas Wolfgruber, Li-Yung Lui, Peter Sadowski, and Steven R Cummings. Deep learning predicts all-cause mortality from longitudinal total-body dxa imaging. *Communications medicine*, 2(1):1–12, 2022.

[5] **Leong Lambert T**, Serghei Malkov, Karen Drukker, Bethany L Niell, Peter Sadowski, Thomas Wolfgruber, Heather I Greenwood, Bonnie N Joe, Karla Kerlikowske, Maryellen L Giger, et al. Dual-energy three-compartment breast imaging for compositional biomarkers to improve detection of malignant lesions. *Communications Medicine*, 1(1):1–11, 2021.

[6] Xun Zhu, Thomas K Wolfgruber, **Leong Lambert**, Matthew Jensen, Christopher Scott, Stacey Winham, Peter Sadowski, Celine Vachon, Karla Kerlikowske, and John A Shepherd. Deep learning predicts interval and screening-detected cancer from screening mammograms: a case-case-control study in 6369 women. *Radiology*, 301(3), 2021.

Patents

Anthropometry by Two-Dimensional Radiographic Imaging. US Patent Application 18/711,165, filed 23 Jan 2025. Shepherd J.A.; Leong L.T.L.K.; Kelly T.L.

Projects

- Biomarker Statistics PyPI Package** | *Model Evaluation, Biostatistics, Matplotlib, Unit Testing*
- Developed and deployed an open-source Python package on PyPI (4.8k downloads in 18 mo), facilitating advanced biomarker evaluation via comprehensive classification metrics and visualizations
 - Emphasized clean, maintainable code, extensive documentation, rigorous unit testing, and adherence to software engineering best practices